

Homework (Carolina Reaper)

- Q1.** Simplify the expression: $\sqrt{64}$
(A) 8
(B) 16
(C) 32
(D) 4
(E) 6
- Q2.** A music producer wants to create a rhythm pattern with a frequency of $\sqrt{121}$. What is the simplified frequency?
(A) 10
(B) 11
(C) 12
(D) 9
(E) 8
- Q3.** An artist wants to create a canvas with a size of $\sqrt{256} \times \sqrt{256}$ inches. What are the dimensions of the canvas?
(A) 16 x 16
(B) 8 x 8
(C) 4 x 4
(D) 2 x 2
(E) 1 x 1
- Q4.** Which expression is equivalent to $\sqrt[3]{8}$?
(A) 2
(B) 2^3
(C) $8^{1/3}$
(D) $8^{2/3}$
(E) $8^{3/2}$
- Q5.** A musician wants to play a song with a tempo of $\sqrt{100}$. What is the simplified tempo?
(A) 5
(B) 10
(C) 20
(D) 50
(E) 100
- Q6.** Simplify the expression: $\sqrt[3]{27}$
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q7.** An artist wants to create a sculpture with a height of $\sqrt{400}$ inches. What is the simplified height?
(A) 10
(B) 20
(C) 30
(D) 40
(E) 50
- Q8.** Which expression is equivalent to $\sqrt[4]{16}$?
(A) 2
(B) $2^{1/2}$
(C) $2^{1/4}$
(D) $16^{1/4}$
(E) $16^{1/2}$

Open Ended Questions (FRQ)

Q9. Simplify the expression: $\sqrt{256} \cdot \sqrt[3]{64}$

Q10. Convert the radical expression to a rational exponent form: $\sqrt[4]{x^3}$

Chef's Tips

1

- Tip 1: Encourage students to use a calculator to check their answers.
- Tip 2: Provide extra support to students who struggle with simplifying radicals.
- Tip 3: Emphasize the importance of converting between radical and rational exponent forms.